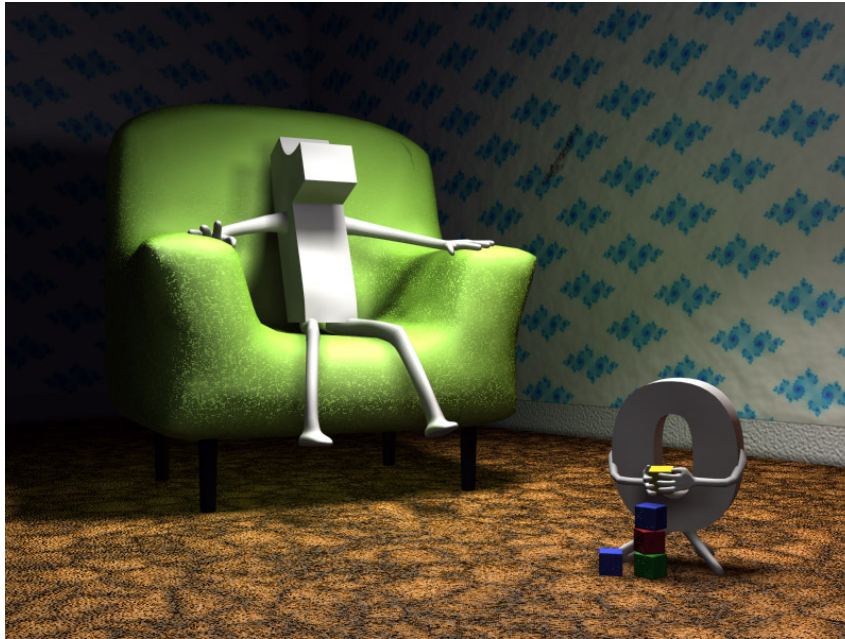


Chapter 1. INTEGERS

1. 3 Integers modulo n

1.3.2 Solving linear equations and systems in $\mathbb{Z}_n$



Today we will learn how to solve a linear system of equations, with different modulus.

Our goal today is to learn how to solve linear systems of the form

$$\left. \begin{array}{l} \bar{x} = \bar{r} \text{ in } \mathbb{Z}_n \\ \bar{x} = \bar{s} \text{ in } \mathbb{Z}_m \end{array} \right\} \Leftrightarrow \left\{ \begin{array}{l} x \equiv r \pmod{n} \\ x \equiv s \pmod{m} \end{array} \right\}$$

where m and n are coprime.

LEMMA 1.53. Let a, b, c be integers such that a, b are relatively prime, and $a|c, b|c$. Then $ab|c$.

Proof. By Bézout's Lemma we have that

$$1 = as + bt, \text{ for some integers } s \text{ and } t.$$

$$a|c \Rightarrow c = ak_1, \text{ for some } k_1 \in \mathbb{Z}$$

$$b|c \Rightarrow c = bk_2, \text{ for some } k_2 \in \mathbb{Z}$$

$$c = 1 \cdot c = (as + bt)c = (as)c + (bt)c$$

$$= (as)(bk_2) + (bt)(ak_1) = \underline{(ab)(sk_2 + tk_1)}$$

$$\Rightarrow \underline{ab|c} \quad \blacksquare$$

CLASSICAL CHINESE REMAINDER THEOREM. Let m, n, s, t be integers with $m \geq 2$ and $n \geq 2$. Assume that m and n are **relatively prime**. Then the simultaneous congruences:

$$x \equiv s \pmod{m}$$

$$x \equiv t \pmod{n}$$

have a common solution. Moreover, any two solutions are congruent modulo mn .

Proof. Every solution of $x \equiv s \pmod{m}$ is of the form $x = s + Km$, for some $K \in \mathbb{Z}$.

$$(m \mid (x-s) \Leftrightarrow x-s = Km \text{ for } K \in \mathbb{Z})$$

Goal. To prove that there exists $K \in \mathbb{Z}$

such that $s + Km \equiv t \pmod{n}$, or equivalently

$Km \equiv t - s \pmod{n}$. Because $\gcd(m, n) = 1$, we can find K .

Suppose that x and y are two common solutions.

$$\left. \begin{array}{l} x \equiv s \pmod{m} \Rightarrow m \mid (x-s) \\ y \equiv s \pmod{m} \Rightarrow m \mid (y-s) \end{array} \right\} \Rightarrow m \mid (x-y)$$

$n \mid (x-y)$. By the previous lemma $mn \mid (x-y)$

$$\Rightarrow x \equiv y \pmod{mn}. \quad \square$$

Example 1.54. Find all the solution to the simultaneous congruences:

$$x \equiv 7 \pmod{8}$$

$$x \equiv 11 \pmod{15}$$

Notice that $\gcd(8, 15) = 1$, so a common solution exists.

Any solution of $x \equiv 7 \pmod{8}$ is of the form $x = 7 + 8K$, for $K \in \mathbb{Z}$.

$$\text{Solve } 7 + 8K \equiv 11 \pmod{15}$$

$$8K \equiv 4 \pmod{15} \quad \bar{8}K = \bar{4} \text{ in } \mathbb{Z}_{15}$$

Find the inverse of $\bar{8}$ in $\mathbb{Z}_{15}$, which is $\bar{2}$.

$$\underline{K \equiv 8 \pmod{15}}$$

$$\underline{x = 7 + 8 \cdot 8 = 71} \text{ is a common solution.}$$

Any two solutions are congruent modulo $15 \cdot 8 = 120$. Thus, all the common solutions are of the form

$$71 + 120n, \quad n \in \mathbb{Z}$$
